

# Cocktail-Party Communication from a Display to a Synchronized Camera

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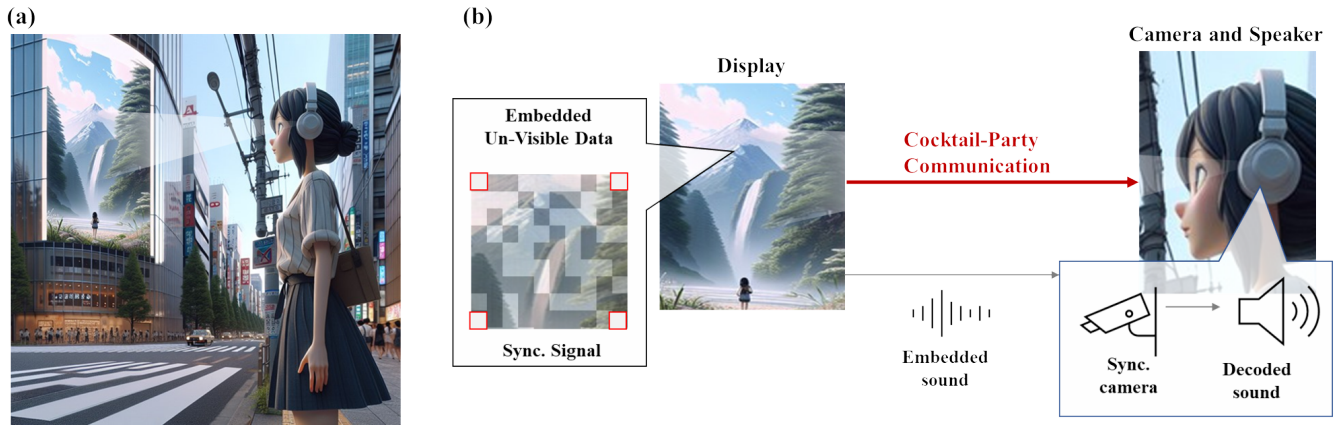


Figure 1: (a) Concept art of CPC, (b) System diagram of CPC.

## CCS Concepts

• Information systems → Multimedia content creation.

## Keywords

optical camera communication, display-camera communication, display-camera synchronization

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## 1 Introduction

Cocktail party effect means a psychologically selective listening of speech [Broadbent 1958]. For example, in a noisy place such as a party, it is like an automatically intensive listening of one's name

and interesting stories. In this paper, Optical Camera Communication (OCC) using a display and a camera realizes the Cocktail-Party Communication (CPC), which is a sound system simulating this effect.

OCC is a type of visible light communication that uses optical signals. It uses light sources including a display or a LED as a transmitter and a camera as a receiver. It is a technology to perform data communication by blinking of light and color information [et al. 2017]. Generally, transmission and reception by OCC are performed asynchronously [et. al. 2021]. In this paper, implementing a camera synchronized with video enables a cocktail party-like selective listening. The synchronized camera realizes stabilization and high data rate of the data communication and enables audio data transmission.

The CPC converts the sound source into optical signals, then demodulates the signals to reproduce audio at the receiving camera. Unlike 10 meters distance limit for Bluetooth, CPC can receive signals from a distant display as a transmitter. As a result, CPC allows us to clearly hear a sound in environments such as urban street displays. It also allows us to clearly hear a particular sound source, a sound corresponding to a video even in environments where multiple sound sources are mixed. (Figure 1(a))

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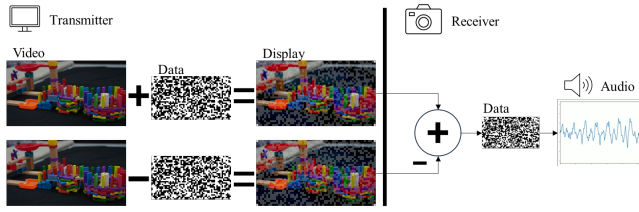


Figure 2: Embedding audio data into video.

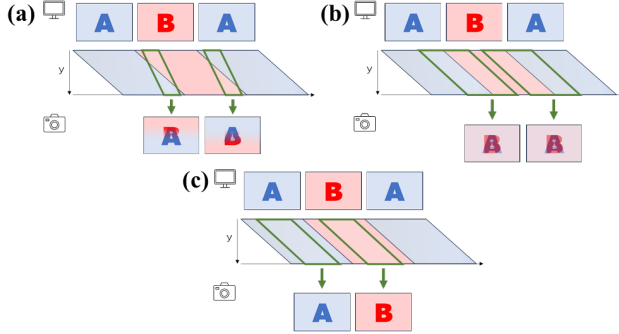


Figure 3: Relationship between display scanning speed and camera shutter speed.

## 2 Approach

Figure 1(b) shows the CPC system. This system reproduces sounds through speakers by pointing a camera at the display. This system embeds audio data into the video so as not to be perceived by the naked eye and displays the video on the display. Then, capturing video images using a camera with a synchronization function acquires frames with embedded audio data. Decoding the data allows us to reproduce the audio data.

## 3 Implementation

Figure 2 shows how to embed audio data in the video. The video is divided into frames, and audio data are added to luminance intensity of odd-numbered frames and subtracted from that of even-numbered frames. Displaying alternately each frame processed by addition and subtraction on the display enables to create a video with embedded audio data that cannot be perceived by the naked eye. To the human eye, displayed images appear to be continuous, however, the camera perceives that the audio data are embedded.

Further, synchronization signals are embedded in the video. When the display and the camera operate at the same frame rate and operate asynchronously, the camera captures images mixed with the preceding and following frames on the display. Our method arranges blinking synchronization signals in the corners of the display. The method calculates the timing information from the blink information and synchronizes the camera with the display timing.

When the display scanning speed and the camera shutter speed are different as shown in Figure 3(a), the images at the top or bottom can become mixed with adjacent images. To achieve the same speed between the display and the camera, first, the synchronization

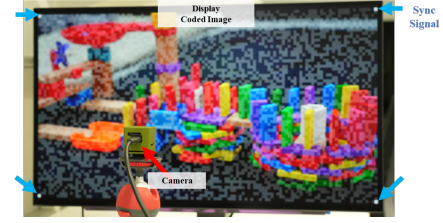


Figure 4: Experimental setup.

signals placed on the corners of the display need to be acquired. Then, the rolling shutter speed of the camera is adjusted using the timing difference information.

As Figure 3(b) shows, when the display timing is different from the image capture timing of the camera, the images of the two frames are mixed. Mixing amounts may vary due to subtle jitters. This case also needs the timing information obtained from the synchronization signals as well. The time per one frame of the camera is temporarily extended or shortened based on the information. This operation synchronizes the display and the camera.

Figure 3(c) shows that the display and the camera are synchronized. This state allows the camera to accurately capture the image of one frame on the display. Synchronization enables to receive signals without any crosstalk between frames. It also achieves stable reception and faster transmission rates.

## 4 Results

Figure 4 shows the state of the implemented system. We operated an OLED display as a transmitter at a rate of 120 fps. We embedded the audio data in the video along with the synchronization signals and displayed the video on the screen. As a receiver, we used Sony IMX577, a rolling shutter image sensor. The frame rate was set to 120 fps and the resolution was set to 1080p. As shown in Figure 3, the rolling shutter speed and frame length were controlled for synchronization. We confirmed that the sound of 8 bit/8 kHz maximum was able to be transmitted under this environment.

## 5 Conclusion

This paper proposes a CPC sound system using a technology to create the communication like the cocktail party effect. We implemented it using OCC and confirmed that the sound of 8 bit/8 kHz maximum was able to be transmitted. We will focus on speeding up code position detection and tracking. We believe that tackling these challenges will lead to faster and more stable communications. By applying the same function, it is also possible to embed video instead of audio. As future works, we would like to propose methods for realizing and implementing this video embedding technology as well.

## References

- D.E. Broadbent. 1958. *Perception and Communication*. Pergamon Press. Pergamon Press.
- J. Klein. et. al. 2021. Investigations on Temporal Sampling and Patternless Frame Recovery for Asynchronous Display-Camera Communication. In *IEEE Trans. on Circuits and System for Video Technology*, Vol. 2. 4004–4015.
- N.T. Le et al. 2017. A survey of design and implementation for optical camera communication. Signal Processing. In *Image Communication*, Vol. 53. 95–109.